

SIM1002 Calculus I
Tutorial 7
Applications of Derivatives

Exercise 4.1

In Exercises 1-2, sketch the graph of each function and determine whether the function has any absolute extreme values on its domain.

1.
$$h(x) = \begin{cases} \frac{1}{x}, & -1 \leq x < 0 \\ \sqrt{x}, & 0 \leq x \leq 4 \end{cases}$$

2.
$$f(x) = \begin{cases} x + 1, & -1 \leq x < 0 \\ \cos x, & 0 < x \leq \frac{\pi}{2} \end{cases}$$

In Exercises 3-6, find absolute maximum and minimum values of each function on the given interval. Then graph the function. Identify the points on the graph where the absolute extrema occur, and include their coordinates.

3.
$$F(x) = -\frac{1}{x}, -2 \leq x \leq -1$$

4.
$$h(x) = -3x^{\frac{2}{3}}, -1 \leq x \leq 1$$

5.
$$g(x) = \sec x, -\frac{\pi}{3} \leq x \leq \frac{\pi}{6}$$

6.
$$f(t) = |t - 5|, 4 \leq t \leq 7$$

In Exercises 7 and 8, find the function's absolute maximum and minimum values and say where they occur.

7.
$$f(x) = x^{\frac{5}{3}}, -1 \leq x \leq 8$$

8.
$$h(\theta) = 3\theta^{\frac{2}{3}}, -27 \leq x \leq 8$$

Finding Critical Points.

In Exercises 9-11, determine all critical points and all domain endpoint for each function.

9.
$$f(x) = \frac{x^2}{x-2}$$

10.
$$g(x) = \sqrt{2x - x^2}$$

11.
$$y = x - 3x^{\frac{2}{3}}$$

12. If an even function $f(x)$ has a local maximum value at $x = c$, can anything be said about the value of f at $x = -c$? Give reasons for your answer.
13. Consider the cubic function

$$f(x) = ax^3 + bx^2 + cx + d.$$

- a) Show that f can have 0, 1, or 2 critical points. Give examples and graphs to support your argument.
- b) How many local extreme values can f have?

Exercises 4.2

Checking the Mean Value Theorem

Find the value or values of c that satisfy the equation

$$\frac{f(b) - f(a)}{b - a} = f'(c)$$

In conclusion of the Mean Value Theorem for the functions and intervals in Exercises 1-2.

1. $f(x) = \sqrt{x - 1}$, $[1, 3]$

2. $f(x) = \begin{cases} x^3, & -2 \leq x \leq 0 \\ x^2, & 0 < x \leq 2 \end{cases}$

Which of the functions in Exercises 3-5 satisfy the hypotheses of the Mean Value Theorem on the given interval, and which do not? Give reasons for your answer.

3. $f(x) = \begin{cases} \frac{\sin x}{x}, & -\pi \leq x < 0 \\ 0, & x = 0 \end{cases}$

4. $f(x) = \begin{cases} 2x - 3, & 0 \leq x \leq 2 \\ 6x - x^2 - 7, & 2 < x \leq 3 \end{cases}$

5. For what values of a , m and b does the function

$$f(x) = \begin{cases} 3, & x = 0 \\ -x^2 + 3x + a, & 0 < x < 1 \\ mx + b, & 1 \leq x \leq 2 \end{cases}$$

satisfy the hypotheses of the Mean Value Theorem on the interval $[0, 2]$?

Roots (zeros)

6. Suppose that f'' continuous on $[a, b]$ and that f has three zeros in the interval. Show that f'' has at least one zero in (a, b) . Generalize this result.
7. Show that if $f'' > 0$ throughout an interval $[a, b]$, then f' has at most one zero in $[a, b]$. What if $f'' < 0$ throughout $[a, b]$ instead?
8. Show that a cubic polynomial can have at most three real zeros.

Show that the functions in Exercises 9-10 have exactly one zero in the given interval.

9. $g(t) = \frac{1}{1-t} + \sqrt{1+t} - 3.1$, $(-1, 1)$
10. $r(\theta) = 2\theta - \cos^2 \theta + \sqrt{2}$, $(-\infty, \infty)$
11. What can be said about functions whose derivatives are constant? Given reasons for your answer.

In Exercises 12-13, find all possible functions with the given derivative.

12. a) $y' = \frac{1}{2\sqrt{x}}$ b) $y' = \frac{1}{\sqrt{x}}$ c) $y' = 4x - \frac{1}{\sqrt{x}}$
13. a) $y' = \sec^2 \theta$ b) $y' = \sqrt{\theta}$ c) $y' = \sqrt{\theta} - \sec^2 \theta$